

Boss Challenge — Paper 23 (Class 7)

Light | Lines & Angles | Respiration in Organisms | Integers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Light from a torch travels from the bulb to the wall along a path that is:
(A) a zig-zag
(B) a straight line
(C) a wavy curve
(D) a circle
2. Two rays that start from the same end-point together form a figure called a/an:
(A) angle
(B) line segment
(C) circle
(D) triangle
3. The process by which living things break down food to release the energy stored in it is called:
(A) germination
(B) transpiration
(C) digestion
(D) respiration
4. The collection of numbers ... -3, -2, -1, 0, 1, 2, 3 ... which includes negatives, zero and positives, is called the:
(A) integers
(B) only the positive numbers
(C) decimals
(D) fractions
5. The bouncing back of light when it falls on a shiny surface such as a mirror is called:
(A) refraction
(B) absorption
(C) reflection
(D) dispersion
6. An angle that measures exactly 90 degrees is called a:
(A) reflex angle
(B) straight angle
(C) acute angle
(D) right angle
7. The gas that our body takes IN from the air during breathing, to help release energy, is:
(A) carbon dioxide
(B) oxygen
(C) nitrogen
(D) hydrogen

8. The value of $(-5) + 3$ is:

- (A) 2
- (B) -2
- (C) -8
- (D) 8

9. When a ray of light reflects from a plane mirror, the angle of incidence is always:

- (A) always ninety degrees
- (B) equal to the angle of reflection
- (C) twice the angle of reflection
- (D) always zero

10. An angle that is greater than 0 degrees but less than 90 degrees is called:

- (A) an acute angle
- (B) a straight angle
- (C) an obtuse angle
- (D) a right angle

11. The waste gas that our body gives OUT when we breathe, and which turns lime water milky, is:

- (A) helium
- (B) hydrogen
- (C) carbon dioxide
- (D) oxygen

12. The value of $(-7) + (-2)$ is:

- (A) 5
- (B) -5
- (C) -9
- (D) 9

13. The image of your face in a flat bathroom mirror is described as:

- (A) virtual, erect and the same size
- (B) larger than your face
- (C) real and upside down
- (D) smaller than your face

14. An angle that is greater than 90 degrees but less than 180 degrees is called:

- (A) a reflex angle
- (B) an acute angle
- (C) an obtuse angle
- (D) a right angle

15. Respiration that uses oxygen to break down food, as in humans, is called:

- (A) transpiration
- (B) anaerobic respiration
- (C) aerobic respiration
- (D) digestion

16. The value of $8 + (-12)$ is:

- (A) 4
- (B) -20
- (C) 20
- (D) -4

17. When you raise your right hand before a plane mirror, the image seems to raise the hand on your left. This swapping of left and right is called:

- (A) magnification
- (B) refraction
- (C) dispersion
- (D) lateral inversion

18. Two angles whose measures add up to exactly 90 degrees are called:

- (A) vertically opposite angles
- (B) complementary angles
- (C) supplementary angles
- (D) reflex angles

19. When muscles work very hard during exercise and run short of oxygen, they release energy WITHOUT oxygen. This is called:

- (A) photosynthesis
- (B) anaerobic respiration
- (C) aerobic respiration
- (D) evaporation

20. The value of $(-6) - (-4)$ is:

- (A) 10
- (B) -10
- (C) 2
- (D) -2

21. In a plane mirror, an object placed 30 cm in front of the mirror forms its image at:

- (A) 30 cm in front of the mirror
- (B) 15 cm behind the mirror
- (C) 30 cm behind the mirror
- (D) 60 cm behind the mirror

22. Two angles whose measures add up to exactly 180 degrees are called:

- (A) complementary angles
- (B) supplementary angles
- (C) acute angles
- (D) equal angles

23. In human beings, the main organs in which the exchange of gases with the air takes place are the:

- (A) kidneys
- (B) stomach
- (C) lungs
- (D) liver

24. The value of $(-4) \times (-3)$ is:

- (A) 12
- (B) -7
- (C) 7
- (D) -12

25. A mirror whose reflecting surface curves inwards, like the inside of a spoon, is a:

- (A) lens
- (B) plane mirror
- (C) convex mirror
- (D) concave mirror

26. The complement of an angle of 30 degrees is an angle of:

- (A) 70 degrees
- (B) 150 degrees
- (C) 60 degrees
- (D) 30 degrees

27. The dome-shaped sheet of muscle below the lungs that moves down to help us breathe in is the:

- (A) the ribs
- (B) the heart
- (C) diaphragm
- (D) the windpipe

28. The value of $(-5) \times 4$ is:

- (A) -9
- (B) -1
- (C) 20
- (D) -20

29. The mirror used as a vehicle's rear-view mirror, because it gives a small upright image and a wide view, is a:

- (A) magnifying glass
- (B) concave mirror
- (C) plane mirror
- (D) convex mirror

- 30.** The supplement of an angle of 110 degrees is an angle of:
- (A) 250 degrees
 - (B) 80 degrees
 - (C) 70 degrees
 - (D) 90 degrees
- 31.** The number of times a healthy person breathes in and out in one minute while resting is closest to:
- (A) about 2 times
 - (B) about 100 times
 - (C) about 500 times
 - (D) about 15 to 18 times
- 32.** The value of $(-20) / 5$ is:
- (A) -100
 - (B) -15
 - (C) 4
 - (D) -4
- 33.** A transparent piece of glass, thicker in the middle than at the edges, that bends light to a point is a:
- (A) convex lens
 - (B) concave lens
 - (C) convex mirror
 - (D) plane mirror
- 34.** When two straight lines cross, the pair of angles that lie exactly opposite each other across the crossing point are:
- (A) always equal
 - (B) always different
 - (C) always adding to 90 degrees
 - (D) always 90 degrees
- 35.** When yeast respires without oxygen, as in making bread and dough rise, it produces carbon dioxide and:
- (A) alcohol
 - (B) pure water only
 - (C) oxygen
 - (D) salt
- 36.** Dividing one negative by another, the value of $(-36) / (-9)$ works out to:
- (A) -4
 - (B) 324
 - (C) -45
 - (D) 4

37. White sunlight passing through a glass prism spreads out into a band of seven colours. This splitting of white light is called:

- (A) absorption
- (B) dispersion
- (C) lateral inversion
- (D) reflection

38. A ray of light strikes a mirror with an angle of incidence of 40° . Using the law that the reflected angle equals it, the angle of reflection is:

- (A) 40 degrees
- (B) 20 degrees
- (C) 50 degrees
- (D) 80 degrees

39. The painful muscle cramp you sometimes feel after very hard exercise is caused by a build-up of:

- (A) lactic acid
- (B) pure water
- (C) carbon dioxide stored in the bones
- (D) extra oxygen

40. On a cold morning the temperature falls from 5°C to -3°C . The change in temperature is:

- (A) a rise of 8°C
- (B) a drop of 2°C
- (C) a drop of 15°C
- (D) a drop of 8°C

41. The correct order of the seven colours of the rainbow, from the inner to the outer edge, begins with violet and ends with:

- (A) red
- (B) violet
- (C) white
- (D) black

42. A light ray hits a mirror so that its angle of incidence is 30° . The total angle between the incident ray and the reflected ray is:

- (A) 90 degrees
- (B) 30 degrees
- (C) 60 degrees
- (D) 15 degrees

43. The pipe that carries air from the nose and throat down towards the lungs is called the windpipe, or:

- (A) xylem
- (B) oesophagus
- (C) artery
- (D) trachea

44. During the day the temperature rises from -4°C to 6°C . The rise in temperature is:
- (A) -10°C
 - (B) 10°C
 - (C) 24°C
 - (D) 2°C
45. A rainbow appears in the sky after rain because tiny water droplets in the air act like prisms and cause:
- (A) the Sun changing colour
 - (B) dispersion of sunlight
 - (C) absorption of sunlight
 - (D) reflection of moonlight
46. Two adjacent angles that together lie along a straight line form a linear pair, and they always add up to:
- (A) 60 degrees
 - (B) 360 degrees
 - (C) 90 degrees
 - (D) 180 degrees
47. Inside the lungs, the exchange of oxygen and carbon dioxide actually takes place in tiny balloon-like sacs called:
- (A) kidneys
 - (B) stomata
 - (C) nostrils
 - (D) alveoli
48. The additive inverse (opposite) of the integer -7 is:
- (A) 0
 - (B) -7
 - (C) $1/7$
 - (D) 7
49. An image that can be caught and shown on a screen, like the picture from a cinema projector, is called a:
- (A) shadow
 - (B) real image
 - (C) virtual image
 - (D) mirror image
50. Two straight lines in the same plane that never meet, however far they are extended, are called:
- (A) perpendicular lines
 - (B) intersecting lines
 - (C) curved lines
 - (D) parallel lines

51. Fish are able to breathe under water because they take in oxygen dissolved in the water through their:
- (A) leaves
 - (B) skin only
 - (C) gills
 - (D) lungs
52. Which of these two integers is the greater: -2 or -8?
- (A) -2
 - (B) they are equal
 - (C) -8
 - (D) neither, both are zero
53. To measure the angle of incidence of a ray striking a mirror, the angle is taken between the ray and:
- (A) the normal
 - (B) the edge of the table
 - (C) the reflected ray
 - (D) the mirror surface itself
54. A straight line that cuts across two or more other lines is called a:
- (A) transversal
 - (B) perpendicular
 - (C) diagonal
 - (D) bisector
55. Insects such as the cockroach take air into their bodies through tiny holes along their sides called:
- (A) nostrils
 - (B) stomata
 - (C) spiracles
 - (D) gills
56. On the number line, the integer -3 lies:
- (A) to the right of 0
 - (B) to the left of 0
 - (C) exactly on 0
 - (D) to the right of 5
57. We are able to see a non-luminous object, such as a book, because the book:
- (A) absorbs all the light
 - (B) makes its own light
 - (C) bends light through itself
 - (D) reflects light into our eyes

58. When a transversal crosses two parallel lines, each pair of corresponding angles is:
- (A) always 90 degrees
 - (B) always different
 - (C) equal
 - (D) adding to 90 degrees
59. Plants also respire, taking in oxygen and giving out carbon dioxide. In the leaves, the gases pass in and out mainly through tiny pores called:
- (A) spiracles
 - (B) stomata
 - (C) roots hairs
 - (D) alveoli
60. When any integer is added to its own additive inverse, the result is always:
- (A) always positive
 - (B) 0
 - (C) 1
 - (D) twice the number
61. A concave mirror is used by a dentist to look at a tooth because, held close, it gives an image that is:
- (A) enlarged and upright
 - (B) coloured like a rainbow
 - (C) smaller and upside down
 - (D) the same size as the tooth
62. When a transversal crosses two parallel lines, a pair of co-interior (allied) angles on the same side adds up to:
- (A) 360 degrees
 - (B) 180 degrees
 - (C) 90 degrees
 - (D) equal, whatever the size
63. An earthworm has no lungs or gills. It takes in oxygen through its:
- (A) moist skin
 - (B) a single lung
 - (C) feathers
 - (D) dry scales
64. A deep-freezer is at -18°C . If it warms up by 5°C after the door is left open, its new temperature is:
- (A) -13°C
 - (B) -23°C
 - (C) -18°C
 - (D) 13°C

65. Light falling straight onto a mirror, exactly along the normal, reflects back along the same path. Its angle of incidence is:
- (A) zero degrees
 - (B) ninety degrees
 - (C) one hundred eighty degrees
 - (D) forty-five degrees
66. In a reflection experiment, the normal is drawn perpendicular to the mirror. The angle between the flat mirror surface and the normal is:
- (A) 180 degrees
 - (B) 45 degrees
 - (C) 90 degrees
 - (D) 0 degrees
67. Bubbling the air a person breathes out through clear lime water turns it milky. This simple test shows that exhaled air is rich in:
- (A) nitrogen
 - (B) oxygen
 - (C) carbon dioxide
 - (D) water vapour
68. Multiplying three negative ones together, the value of $(-1) \times (-1) \times (-1)$ comes to:
- (A) 3
 - (B) 1
 - (C) -1
 - (D) -3
69. A concave lens, which is thinner in the middle than at its edges, affects passing light by:
- (A) splitting the rays into colours
 - (B) spreading the rays apart
 - (C) bringing the rays to a point
 - (D) bouncing the rays straight back
70. A light ray makes an angle of 20° with the flat surface of a mirror. Since angles of incidence are measured from the normal, the angle of incidence is:
- (A) 110 degrees
 - (B) 70 degrees
 - (C) 20 degrees
 - (D) 90 degrees
71. When a person starts running fast, their breathing rate compared with when resting will:
- (A) stay exactly the same
 - (B) decrease
 - (C) increase
 - (D) stop altogether

72. The value of $0 \times (-99)$ is:

- (A) -1
- (B) -99
- (C) 99
- (D) 0

73. A flat, smooth, polished surface gives a clear sharp image, while a rough wall does not, because a rough surface scatters light in:

- (A) the colours of the rainbow
- (B) a perfect circle
- (C) many different directions
- (D) one single direction

74. One of two complementary angles is 55 degrees. The other angle must be:

- (A) 45 degrees
- (B) 35 degrees
- (C) 125 degrees
- (D) 55 degrees

75. Word equation for aerobic respiration: glucose + oxygen $\rightarrow$ carbon dioxide + energy + _____. The missing product is:

- (A) oxygen
- (B) water
- (C) nitrogen
- (D) more glucose

76. The value of $4 - 9$ is:

- (A) -13
- (B) 13
- (C) 5
- (D) -5

77. The image formed in a plane mirror is called virtual because the light rays:

- (A) are completely absorbed by the glass
- (B) only appear to come from behind the mirror
- (C) split into seven colours
- (D) actually meet on a screen behind it

78. Two angles form a linear pair. If one of them is 125 degrees, the other is:

- (A) 125 degrees
- (B) 55 degrees
- (C) 65 degrees
- (D) 45 degrees

79. Compared with the air we breathe in, the air we breathe out contains more carbon dioxide and more:

- (A) oxygen
- (B) nitrogen
- (C) dust
- (D) water vapour

80. Multiplying any integer by 1 gives:

- (A) zero
- (B) always a positive number
- (C) the same integer
- (D) double the integer

81. When a beam of light passes from air into a glass slab, it bends. This bending of light on entering a new material is called:

- (A) dispersion
- (B) reflection
- (C) refraction
- (D) transparency

82. When two lines cross, one angle is found to be 65 degrees. The angle vertically opposite to it is:

- (A) 90 degrees
- (B) 65 degrees
- (C) 115 degrees
- (D) 25 degrees

83. Compared with aerobic respiration, anaerobic respiration releases:

- (A) less energy from the same food
- (B) much more energy
- (C) exactly the same energy
- (D) no energy at all

84. At dawn the temperature is -2°C and by noon it is 11°C . The difference between the two temperatures is:

- (A) 9°C
- (B) 13°C
- (C) 11°C
- (D) 22°C

85. A magnifying glass that makes small print look bigger is, in fact, a:

- (A) plane mirror
- (B) concave lens
- (C) glass prism
- (D) convex lens

- 86.** A transversal crosses two parallel lines and one corresponding angle is 80 degrees. Its matching corresponding angle is:
- (A) 90 degrees
 - (B) 100 degrees
 - (C) 10 degrees
 - (D) 80 degrees
- 87.** The roots of a plant also need to respire. This is why plants growing in waterlogged soil may suffer, because the water:
- (A) turns the roots into gills
 - (B) makes the leaves respire faster
 - (C) drives air out of the soil
 - (D) gives the roots extra oxygen
- 88.** Arranged from the smallest to the largest, the integers -5, -1, 0 and 3 begin with:
- (A) -1
 - (B) 0
 - (C) -5
 - (D) 3
- 89.** If a ray of light strikes a plane mirror at an angle of incidence of 50° , the angle of reflection will be:
- (A) 40 degrees
 - (B) 25 degrees
 - (C) 100 degrees
 - (D) 50 degrees
- 90.** An angle whose measure is more than 180 degrees but less than 360 degrees is called:
- (A) a straight angle
 - (B) an obtuse angle
 - (C) a reflex angle
 - (D) a right angle
- 91.** Before air reaches the lungs, the nose helps by warming the air and also:
- (A) adding carbon dioxide
 - (B) filtering out dust and germs
 - (C) removing all the oxygen
 - (D) cooling the air right down
- 92.** The value of $(-3) \times (5 + (-5))$ is:
- (A) -30
 - (B) 30
 - (C) 0
 - (D) -3

93. Objects that give out their own light, such as the Sun and a glowing bulb, are described as:

- (A) luminous
- (B) non-luminous
- (C) transparent
- (D) opaque

94. Two angles are supplementary and one of them measures exactly 90 degrees. The other angle must be:

- (A) 90 degrees
- (B) 0 degrees
- (C) 180 degrees
- (D) 45 degrees

95. All the energy released by respiration in living cells originally comes from the food, which in green plants is first made using the energy of:

- (A) wind
- (B) sunlight
- (C) the soil's heat
- (D) moonlight

96. The only integer that is neither positive nor negative is:

- (A) -1
- (B) there is no such integer
- (C) 1
- (D) 0

97. A convex mirror is preferred over a plane mirror as a shop security or rear-view mirror mainly because it:

- (A) covers a wider area
- (B) splits light into colours
- (C) shows a real image on a screen
- (D) magnifies everything

98. The figure made by a single straight, never-ending path with no thickness and no end-points is called a:

- (A) line
- (B) ray
- (C) line segment
- (D) angle

99. A single-celled organism like Amoeba does not need special breathing organs because it can:

- (A) exchange gases straight through its surface
- (B) hold its breath for hours
- (C) grow tiny lungs when needed
- (D) make its own oxygen inside

100. A diver starts at the surface (0 m), descends to a depth of -15 m, then rises 6 m. The diver's new position is:

- (A) -9 m
- (B) -15 m
- (C) -21 m
- (D) 9 m